

Understanding a Spiking Neuron: A Tutorial on Leaky Integrate-and-Fire Dynamics and Implementation

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SCOPE

This feature article proposes a tutorial on Spiking Neural Networks (SNNs), focusing on the Leaky Integrate-and-Fire (LIF) neuron model. The article presents both the theoretical background and practical implementations of LIF dynamics, including charging, discharging, threshold-triggered resets, and surrogate gradient training. It is aimed at readers seeking a rigorous but accessible introduction to spiking neuron models and their computational behavior.

I. HISTORY, MOTIVATION, AND SIGNIFICANCE OF THE TOPIC

SNNs represent the third generation of neural network models and are inspired by the event-driven communication of biological neurons. Their energy efficiency and temporal processing capabilities make them ideal for edge computing and neuromorphic hardware. Despite the emergence of frameworks like SNN Torch, a gap remains in accessible, mathematically grounded tutorials that elucidate how LIF neurons function and are implemented. This tutorial addresses that gap by combining the understanding of RC circuits, numerical methods and surrogate gradient techniques.

While some literature focuses on biologically detailed models (e.g., Hodgkin-Huxley), these are often computationally prohibitive. The proposed article emphasizes the LIF model for its balance between biological plausibility and computational efficiency. We also distinguish the article from prior work by offering clean and modular implementations suitable for education and integration into larger SNN systems.

II. OUTLINE OF THE PROPOSED FEATURE ARTICLE

- **Introduction**
 - Motivation: energy efficiency, relevance of SNNs.
 - Overview of tutorial structure.
- **Biological and Computational Background**
 - Biological spiking and event-driven computation.
 - LIF neuron as an abstraction of membrane dynamics.
- **The LIF Model: Physical and Mathematical Foundations**
 - Derivation from RC circuits.
 - Charging and discharging behavior.
 - Time constant, threshold, and reset mechanisms.
- **Simplified Model for Implementation**
 - Reducing parameter space.
 - Derivation of membrane potential update in discrete time.
 - Role of decay factor β .
- **Implementation**
 - Code listings for charging/discharging.
 - Full neuron implementation with thresholding.
 - Plotting and interpretation of simulation results.
- **Training Spiking Neurons**
 - Surrogate gradients and differentiable approximations.
 - Use of HardTanh and derivative.
 - End-to-end forward and backward pass in code.
- **Conclusion and Further Directions**

- Summary of LIF benefits and limitations.
- Extensions to full SNN architectures.
- Open problems and next steps.

III. PROJECTED SUBMISSION DATE

Within two months after acceptance of this proposal.

AUTHORS

André Furlan holds a Master's degree in Computer Science from UNESP (Universidade Estadual Paulista), where he also taught. He currently teaches at CEETEPS (Paula Souza Center) and conducts research in artificial intelligence, signal processing, and computational neuroscience. His experience includes systems development and applied research in brain–computer interfaces, EEG and voice signal analysis, and neuromorphic modeling using Spiking Neural Networks (SNNs). His academic background spans applied mathematics, programming, and bio-inspired computing. He is a reviewer for Springer journals and actively contributes to scientific dissemination and technical education in computing.